CS316 Quiz 3: Interpretation Question Guide

How the instructor writes them, model answers, and harder practice (real outputs)

PART 1: HOW THE QUESTIONS ARE WRITTEN

- **Every question sits inside a scenario** (hospital, length of stay). Use the context: days, patients, beds, not just 'the value increases'.
- **Read an output, classify, justify with the number.** Name the shape and cite the number.
- **Decisions must be defended with a concept** (e.g. missing at random vs not at random). The concept word is expected.
- **Coefficients are interpreted in plain words, with units.**
- **Answers are short (2-4 sentences)**: a claim, the evidence, the consequence.
- **Code and interpretation are linked:** you interpret plots made from code you completed.

Answer template: [Claim] + [the number / evidence] + [what it means for this scenario]

PART 2: QUIZ 3 MODEL ANSWERS

Real outputs: missing BMI 14%, BP 9%; skewness 0.296; mean 12.79, median 12.4; IQR bounds [-2.45, 27.55], 2 outliers; intercept -5.694; coefficients age 0.123, bmi 0.169, systolic_bp 0.013, prior_admissions 3.022, bmi_missing -0.249; test MSE 5.598, MAE 1.783, RMSE 2.366; train MSE 4.692; new patient about 15.9 days.

QV.2: Interpret the distribution

(1) length_of_stay is **roughly symmetric, with a slight right skew**. (2) Skewness 0.296 is positive but small (below 0.5): a mild tail of longer stays, consistent with the mean (12.79) being slightly above the median (12.4).

Q1.2: Is it safe to drop rows with missing BMI?

No. BMI is missing more often for very ill or emergency patients, so it is **missing not at random (MNAR)**: the missingness depends on the patients' condition. Dropping those rows would remove the sickest patients and bias the data toward healthier, shorter stays. That is why the quiz keeps them and adds a bmi_missing flag.

Q3.5: Interpret the prior_admissions coefficient (3.022)

Holding the other features constant, each additional prior admission is associated with about **3 more days** in hospital.

Code key

```
missing_pct = df.isnull().mean() * 100          # or df.isnull().sum() / len(df) * 100

df_imp = df.copy()
df_imp["bmi_missing"] = df["bmi"].isnull().astype(int)          # 1 = missing, 0 = measured
df_imp["bmi"] = df_imp["bmi"].fillna(df_imp["bmi"].median())
df_imp["systolic_bp"] = df_imp["systolic_bp"].fillna(df_imp["systolic_bp"].median())

def mse(y_true, y_pred):  return np.mean((y_true - y_pred) ** 2)
def mae(y_true, y_pred):  return np.mean(np.abs(y_true - y_pred))
def rmse(y_true, y_pred): return np.sqrt(mse(y_true, y_pred))

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
test_mse, test_mae, test_rmse = mse(y_test, y_pred), mae(y_test, y_pred), rmse(y_test, y_pred)

new_patient = pd.DataFrame([[68, 31, 145, 2, 0]], columns=features)
new_pred = model.predict(new_patient)     # array -> new_pred[0] is about 15.9 days
```

Watch out: create bmi_missing BEFORE filling bmi; afterwards there are no NaNs left to detect. Values in the list-of-lists row must follow the order of *features*.

PART 3: HARDER PRACTICE QUESTIONS

Set A: Distributions and skew

A1. A second dataset of ICU stays has mean 9.4 days, median 5.0, skewness 2.1. Describe the distribution and say which statistic you would use to fill missing ICU stays.
A2. The box plot of length_of_stay shows Q1 = 8.8, Q3 = 16.3, and 2 points above 27.55. The manager wants to delete those 2 'outliers' before training. Do you agree? Justify in context.
A3. A colleague says 'skewness is 0.296, so I must use the median for everything.' Is that a good conclusion?

Set B: Missing data decisions

B1. Systolic BP is missing for 9% of patients because a blood-pressure machine on Ward 3 was broken for a week. Which type of missingness is this? Is median imputation reasonable?
B2. Why does the quiz create bmi_missing instead of simply filling BMI? What could the model learn from that column?
B3. The coefficient of bmi_missing is -0.249. Interpret it. Given that missing BMI was supposed to signal very ill patients, is this surprising?
B4. A student fills missing BMI with the mean of the whole dataset, including the test set, before splitting. What is the problem?

Set C: Loss metrics

C1. Test MAE = 1.783 and RMSE = 2.366. Explain each in days, and what the gap between them tells you.
C2. Model A: MAE 1.8, RMSE 2.4. Model B: MAE 1.9, RMSE 4.5. The hospital mostly fears badly underestimating long stays. Which model, and why?
C3. Train MSE = 4.692, test MSE = 5.598. Is the model overfitting? Justify.

Set D: Coefficients and predictions

D1. The intercept is -5.694. What does it mean literally, and why shouldn't the hospital interpret it?
D2. Age coefficient 0.123, prior_admissions 3.022. A manager says 'prior admissions matter 25 times more than age.' Is that comparison valid?
D3. The model predicts 15.9 days for the new patient. A nurse asks whether he will stay exactly 15.9 days. Answer using the test RMSE.
D4. In the actual-vs-predicted plot, points for actual stays above 20 days lie mostly BELOW the y = x line. What does that mean, and what is the hospital risk?

PART 4: MODEL ANSWERS

A1. Strongly right-skewed: skewness 2.1 is well above 1, and the mean (9.4) is far above the median (5.0) because a few very long ICU stays pull the mean up. Fill with the median: it represents a typical stay and is not distorted by the tail.
A2. No. These are real long-stay patients, not data errors; the scenario says a minority genuinely stay over 20 days. Deleting them would make the model underestimate exactly the long stays that matter most for bed planning.
A3. No. Skewness 0.296 is small (roughly symmetric), so mean and median are close (12.79 vs 12.4) and the mean is also reasonable. The median becomes necessary when skew or outliers are strong.
B1. MCAR: a broken machine is unrelated to the patients' blood pressure or condition, so median imputation is reasonable. (If Ward 3 were a specific type of ward, e.g. cardiology, it could become MAR.)
B2. Filling erases the fact that BMI was missing, and that fact carries information (very ill or emergency patients). The flag lets the model learn whether those patients have different stay lengths.
B3. Holding other features constant, patients with missing BMI are predicted to stay about 0.25 days less: almost no effect. That is surprising given the story; in this data, missing BMI is not strongly linked to stay length.
B4. Data leakage: test-set information influences the training data, so the test score becomes too optimistic. Compute the fill value from the training set only.
C1. MAE: predictions are off by about 1.8 days on average. RMSE: about 2.4 days, with large errors weighted more. RMSE > MAE shows there are some larger errors, not just uniform small ones.

C2. Model A. Their MAEs are similar, but B's RMSE (4.5) is far above its MAE, so B makes some very large errors: the dangerous long-stay misses. RMSE is the better guide when big errors are costly.
C3. Only mildly: test MSE is somewhat higher than train MSE (5.6 vs 4.7), which is normal. A large gap would signal overfitting (high variance); this gap is small, so the model generalizes reasonably.
D1. Literally: the predicted stay for a patient aged 0 with BMI 0, BP 0 and no prior admissions is -5.7 days. That is impossible and outside the data's range, so it has no real meaning; it only positions the line.
D2. Not directly. Coefficients are per unit and the units differ (+1 year vs +1 admission). A patient 25 years older gets about 3 extra days, the same as one extra admission. Fair comparison needs features on the same scale.
D3. No. 15.9 is the model's best estimate. With a test RMSE of about 2.4 days, actual stays commonly differ by around 2-3 days, more for unusual patients.
D4. For long-stay patients the model underpredicts: it predicts shorter stays than really happen. The hospital may plan too few beds or staff for its sickest patients.